

life cycle research summary

A brand of shoe everyone is familiar with, over a billion pairs being sold since the company's birth: Converse. A staple shoe for generations now, making it big in the 1920's with their Chuck Taylor All Stars. Today Converse is owned by Nike and has been since 2003. This change came with a change in material quality for the Chuck Taylor All Stars, their simple design containing cotton canvas, rubber used in the soles, and aluminum eyelets.

These simple shoes are deceptive in their ecological and social impacts. Cotton farming and harvesting is one of the most invasive, wasteful and damaging processes using up to 125 litres of water per kilogram of cotton fibre, 6% of the world's pesticides, and 16% of all insecticides. Long term exposure causes health issues including impaired memory, insomnia, infertility, and higher risks of cancer (Environmental Justice Foundation, 2025). The most humane way to pick cotton, as the cotton farming industry is notorious for slave labour, is a 'Cotton Stripper', stripping the entire cotton boll along with its stalk, many of the bolls end up being picked before they are ripe and then discarded (Tokens, 2021). The natural rubber used in the soles is 'vulcanized' before use. (ScienceDirect, 2014). Energy intensive heating of the rubber with sulfur and other bonding chemicals create a less sticky, flexible material with a high resistance to degradation that is also difficult to recycle without proper machinery, instead leaching chemicals into the soil and water when discarded. The sulfur extraction process also often releases greenhouse gases such as CO₂ and SO₂. (ECHEMI, 2025). Rubber trees are farmed in Southeast Asia where 90% of global rubber is produced as 14.2 million hectares are occupied by rubber plantations, contributing to soil erosion. With 6 million small farms producing the world's rubber, labour and human rights violations are rampant. Collecting rubber is relatively sustainable, but studies have found that workers exposed to dust from rubber trees experience an increased risk of asthma, skin symptoms, fibrosis, and have reduced spirometric lung function. (Sripaiboonkij et al., 2009)

Nike and Converse do not offer transparent data about each individual company, but the companies together have high waste percentages. 26% of all waste from manufacturing, 32% in product packaging, and 32% in shipping (Converse Inquiry Project, 2018). Converse are primarily made in China, Indonesia, and India and then sent to the U.S. for global distribution. Using the trans-pacific trade route, covering around 11,000km which with two tonnes of freight carried by ship produces 330kg of CO₂ and 45,518kg by air (The Real Environmental Impact of Freight Shipping, 2025). To dodge tariffs Converse use felt in 50% of the sole's material, classifying it as a slipper. This drops the rates from around 37% to a guaranteed rate of 3%.

Pre-2003 consumers reported the wearability of Converse to be around 8 years, without much wear and tear, today Converse typically last between 18 months and 24 months with everyday wear, as Converse switched from 2ply to 1ply canvas (Lorna, 2021). With the soles of most shoes leaving a trail of microplastics with every step. Converse has made efforts in their packaging shifting to minimize the number of inserts used, redesigning to minimize plastic use, with boxes made from recycled paper and soy-based inks that don't cause toxic runoff after disposal (Ross, 2026). Though these efforts are intended to be obvious and marketable without considering deeper issues, evident by their durability and manufacturing conditions. "Converse Renew" is a prime example of this and the companies 'green washing'. This initiative prioritizes the reuse of existing materials over the production of new ones, incorporating recycled plastic bottles, and repurposed denim. Marketed as made from fully reused materials, in reality it's around 35% of the sneaker's total weight. Though a start, they only offer styles/colours visibly different to the original shoes and do not offer facilities to actively recycle their own used products.

redesign statement

My redesign addresses the impacts of materials used in Converse through the substitution of cotton and natural rubber with mycelium fabrics and nitrile cork, offsetting energy consumption and pollutants from before manufacturing to the afterlife.

Mycelium can grow from waste materials like sawdust, glass pieces, and rice hulls offering a stronger alternative, similar to leather but entirely biodegradable without toxins. It's land and energy efficient as it grows densely, can be grown indoors and doesn't require much water. Its structure is fibrous and absorbent, allowing it to easily accept natural dyes and pigments, allowing for a wide range of personalization. Mycelium fabrics are generally lighter and more energy-efficient to ship than cotton due to their lower density, which reduces shipping weight and transport-related carbon emissions. Mycelium fabrics are grown in facilities primarily in the US, allowing its production to take place with stricter labour laws and removing the shipment between manufacturer and distributor.

Nitrile cork is a blend of cork and rubber, cork typically at least 35% of the makeup. It's a renewable, biodegradable, and carbon-negative material harvested by stripping the bark of cork oak trees without harming them. Often, using discarded cork from other products. Cork can also add additional cushioning, save weight, and make the soles more comfortable than traditional rubber soles.

People are more likely to want to consume and invest in things that align with their values. This is called value-driven consumption or impact investing, and people are often willing to pay more if so. A company of this scale implementing a redesign such as this, without challenging the iconic integrity of their original shoes, would pave the way for more recognition into mycelium materials and its beneficial uses.

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